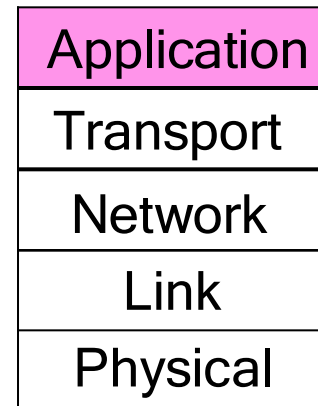


UNIT V-Application Layer

- DNS - Domain Name System
- **Electronic Mail**
- World Wide Web: Architectural Overview
- Content Delivery.

The Application Layer

Uses transport services to build distributed applications



DNS - Domain Name System

- ❑ The DNS resolves high-level human readable names for computers to low-level IP addresses.
- ❑ The essence of DNS is the invention of a hierarchical, domain-based naming scheme and a distributed database system for implementing this naming scheme. It is primarily used for mapping host names to IP addresses .
- ❑ To map a name onto an IP address, an application program calls a library procedure called the resolver, passing it the name as a parameter.
- ❑ The resolver sends a query containing the name to a local DNS server, which looks up the name and returns a response containing the IP address to the resolver, which then returns it to the caller.
- ❑ The query and response messages are sent as UDP packets. Armed with the IP address, the program can then establish a TCP connection with the host or send it UDP packets

The DNS Name Space

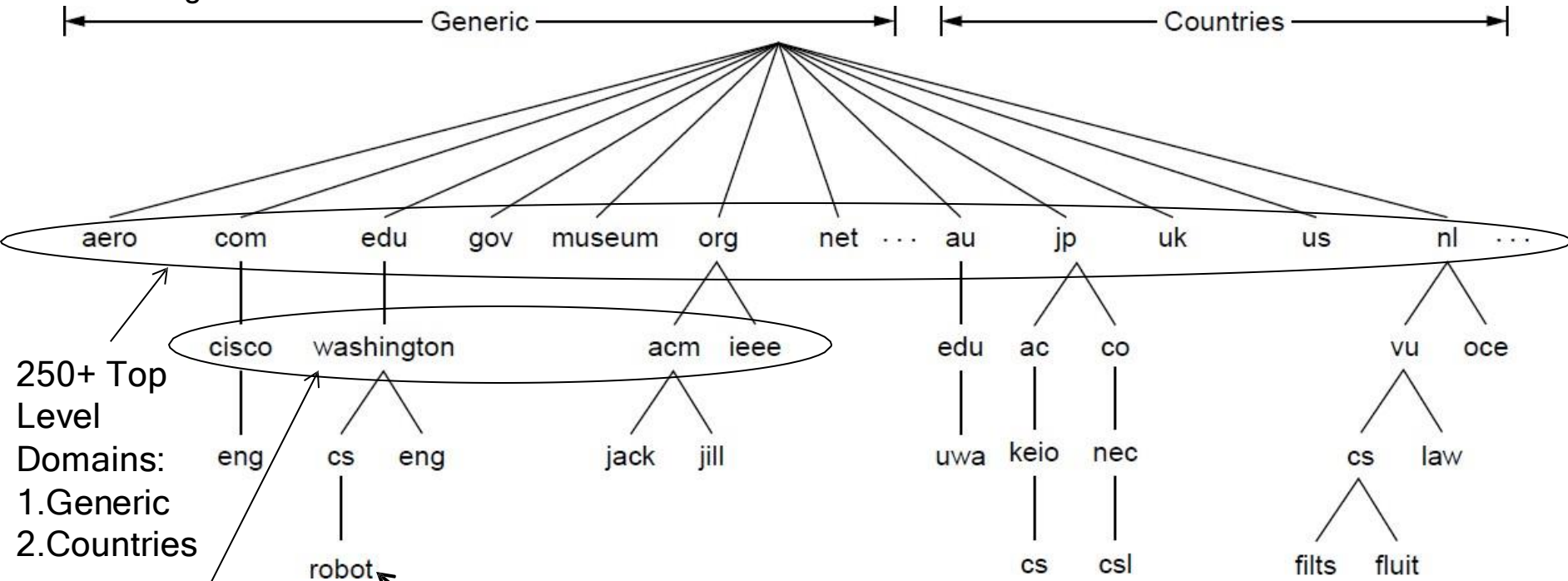
- ❑ For the Internet, the top of the naming hierarchy is managed by an organization called ICANN (Internet Corporation for Assigned Names and Numbers). ICANN was created for this purpose in 1998, as part of the maturing of the Internet to a worldwide, economic concern.
- ❑ Conceptually, the Internet is divided into over 250 top-level domains, where each domain covers many hosts. Each domain is partitioned into subdomains, and these are further partitioned, and so on.
- ❑ All these domains can be represented by a tree, as shown in Fig. 1. The leaves of the tree represent domains that have no subdomains (but do contain machines, of course).
- ❑ A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.
- ❑ The top-level domains come in two flavors: generic and countries. The generic domains, listed in Table 1.
- ❑ The country domains include one entry for every country, as defined in ISO 3166. Internationalized country domain names that use non-Latin alphabets were introduced in 2010. These domains let people name hosts in Arabic, Cyrillic, Chinese, or other languages.

The DNS Name Space (1)

DNS namespace is hierarchical from the root down

- Different parts delegated to different organizations
- Can register under multiple top-level domains
- Absolute domain names always end in a period, relative domain names do not
 - Relative names have to be interpreted in a context to uniquely determine their meaning

Figure:1



The computer *robot.cs.washington.edu*

Generic Second Level
Domains run by Registrars
appointed by ICANN

The DNS Name Space (2)

Table 1

Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
int	International organizations	1988	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No
museum	Museums	2002	Yes
name	People	2002	No
pro	Professionals	2002	Yes
cat	Catalan	2005	Yes
jobs	Employment	2005	Yes
mobi	Mobile devices	2005	Yes
tel	Contact details	2005	Yes
travel	Travel industry	2005	Yes
xxx	Sex industry	2010	No

The Generic top-level domains are controlled by ICANN who appoints registrars to run them

ICANN = Internet Corporation for Assigned Names and Numbers

This one was controversial



To create a new domain, permission is required of the domain in which it will be included.
Naming follows organizational boundaries, not physical networks.

Domain Resource Records

- ❑ Every domain, whether it is a single host or a top-level domain, can have a set of resource records associated with it. These records are the DNS database.
- ❑ For a single host, the most common resource record is just its IP address, but many other kinds of resource records also exist.
- ❑ When a resolver gives a domain name to DNS, what it gets back are the resource records associated with that name.
- ❑ Thus, the primary function of DNS is to map domain names onto resource records. A resource record is a five-tuple. Although they are encoded in binary for efficiency, in most expositions resource records are presented as ASCII text, one line per resource record.

Domain Resource Records (1)

Every domain (all levels) has its own DNS database that is comprised of Resource Records.

Each Resource Record is a 5 tuple: Domain_Name, TTL, Class, Type, Value (page 616)

The key resource records in the namespace are IP addresses (A/AAAA) and name servers (NS), but there are others too (e.g., MX)

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

Domain Resource Records (2)

; Authoritative data for cs.vu.nl

cs.vu.nl.	86400	IN	SOA	star boss (9527,7200,7200,241920,86400)
cs.vu.nl.	86400	IN	MX	1 zephyr
cs.vu.nl.	86400	IN	MX	2 top
cs.vu.nl.	86400	IN	NS	star

← Name server

star	86400	IN	A	130.37.56.205
zephyr	86400	IN	A	130.37.20.10
top	86400	IN	A	130.37.20.11
www	86400	IN	CNAME	star.cs.vu.nl
ftp	86400	IN	CNAME	zephyr.cs.vu.nl

← IP addresses of computers

flits	86400	IN	A	130.37.16.112
flits	86400	IN	A	192.31.231.165
flits	86400	IN	MX	1 flits
flits	86400	IN	MX	2 zephyr
flits	86400	IN	MX	3 top

CNAME and PTR are aliases; CNAME allows the same IP address to have multiple names; PTR enables reverse lookups

rowboat		IN	A	130.37.56.201
		IN	MX	1 rowboat
		IN	MX	2 zephyr

←

little-sister		IN	A	130.37.62.23
---------------	--	----	---	--------------

laserjet		IN	A	192.31.231.216
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A portion of a possible DNS database for cs.vu.nl.

Question: What is cs.vu.nl's web server's name and IP address?

Question: What is a reverse lookup?

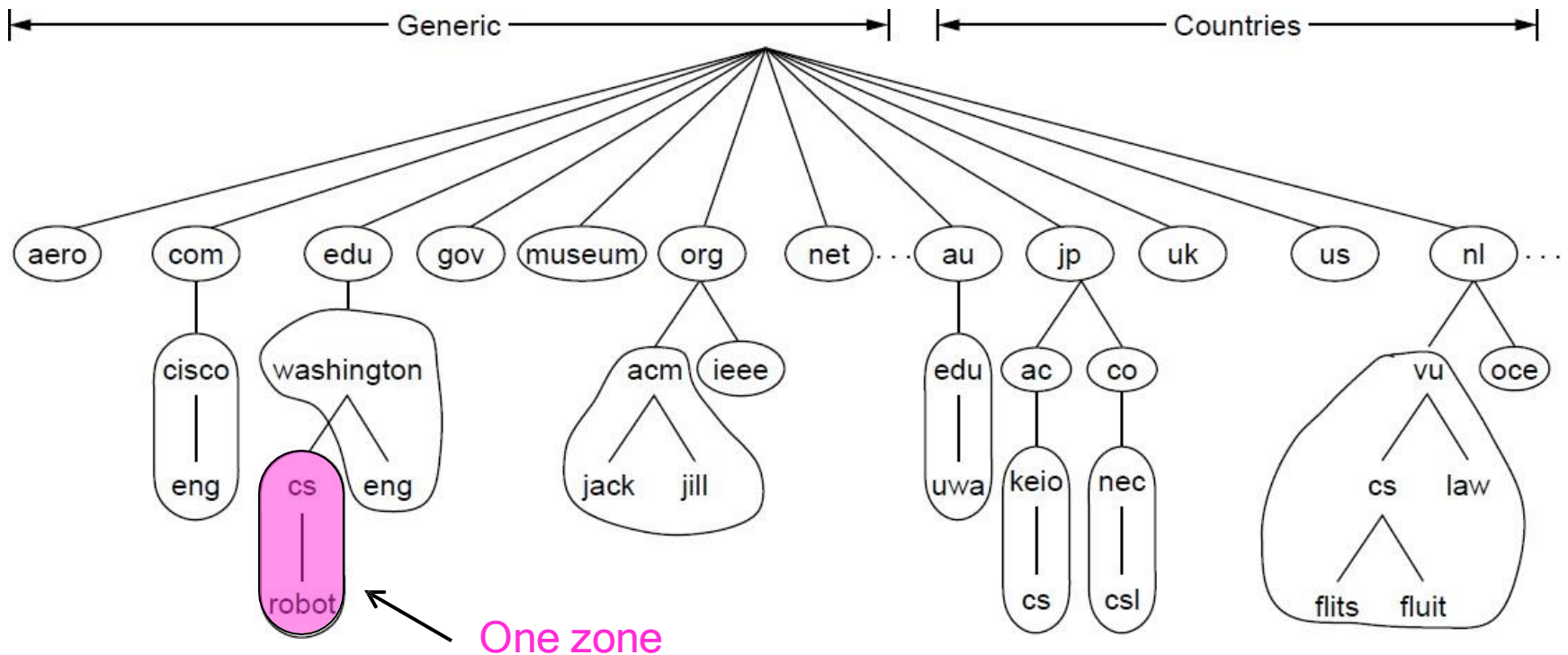
Name Servers

- ❑ In theory at least, a single name server could contain the entire DNS database and respond to all queries about it. In practice, this server would be so overloaded as to be useless. Furthermore, if it ever went down, the entire Internet would be crippled.
- ❑ To avoid the problems associated with having only a single source of information, the DNS name space is divided into nonoverlapping zones.

Name Servers (1)

DNS name space divided into non overlapping zones. Admins determine zone boundaries. Each zone has one or more name servers.

Name servers contain data for portions of the name space called zones (circled).



Name Servers (2)

Finding the IP address for a given hostname is called resolution and is done with the DNS protocol.

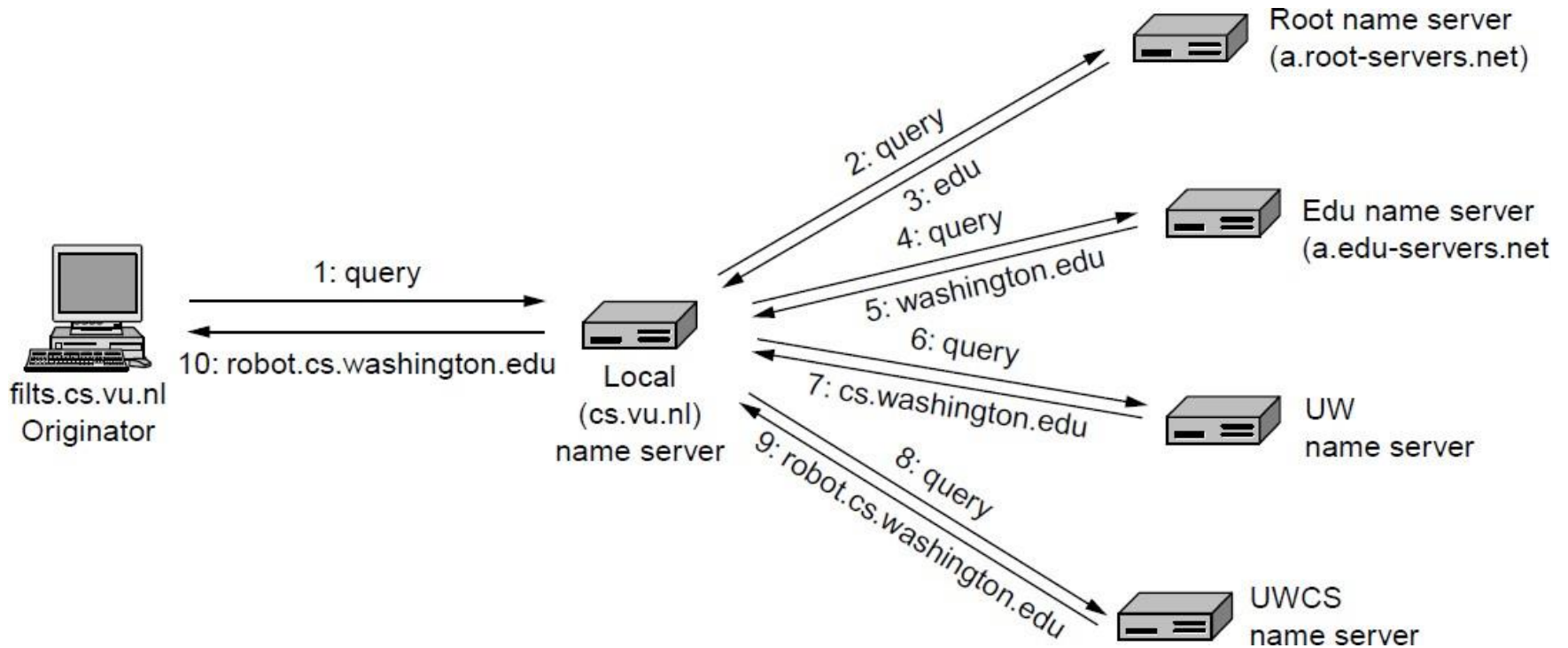
Resolution:

- Computer requests local name server to resolve
- Local name server asks the root name server
- Root returns the name server for a lower zone
- Continue down zones until a name server can answer

DNS protocol:

- Runs on **UDP port 53**, retransmits lost messages
- Caches name server answers for better performance

Name Servers (3)



Example of a computer looking up the IP for a name

Electronic Mail

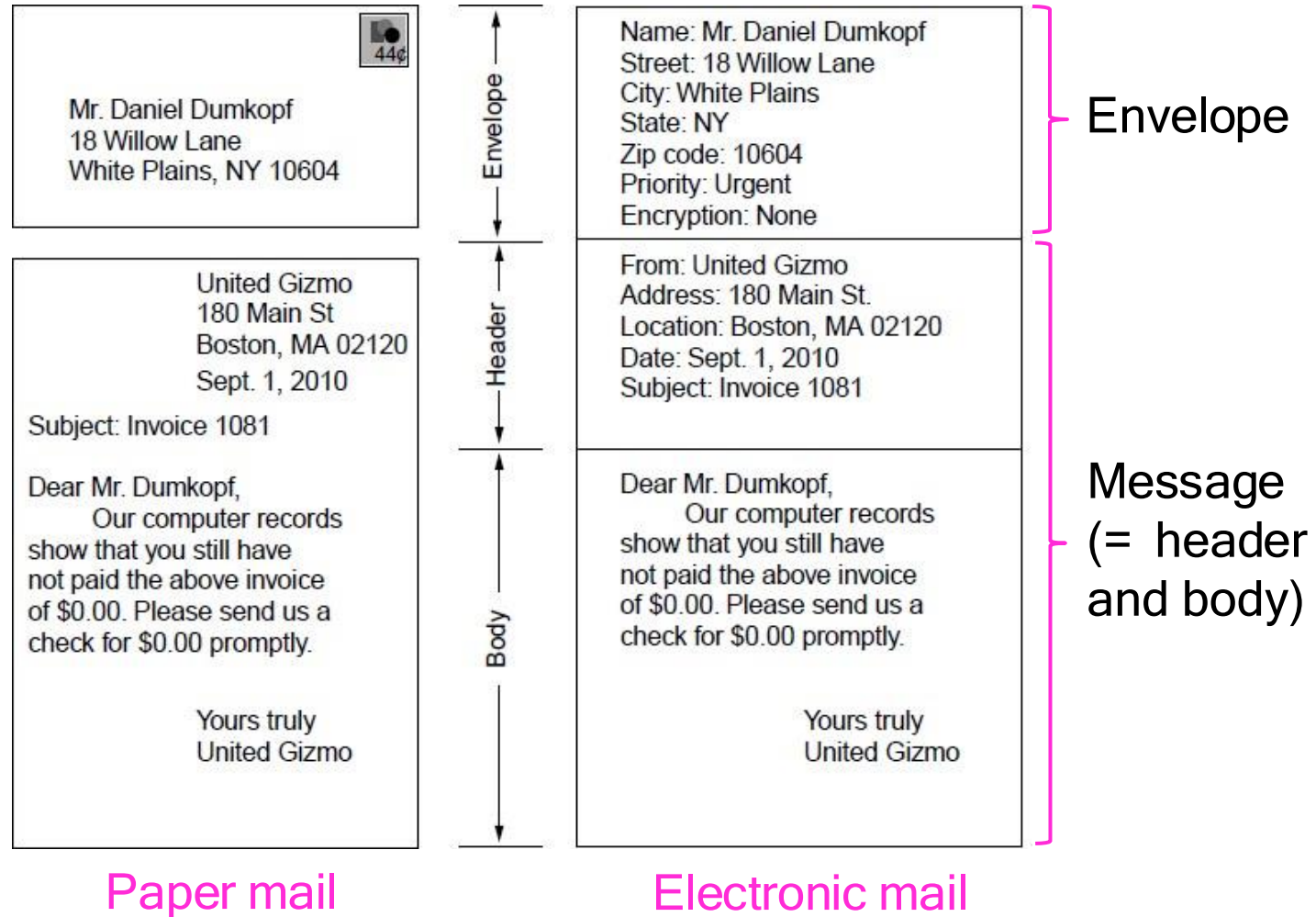
Electronic Mail (e-mail) is one of most widely used services of internet. This service allows an Internet user to send a **message in formatted manner (mail)** to the other Internet user in any part of world.

Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called **sender** and person who receives mail is called **recipient**. It is just like postal mail service.

Electronic Mail

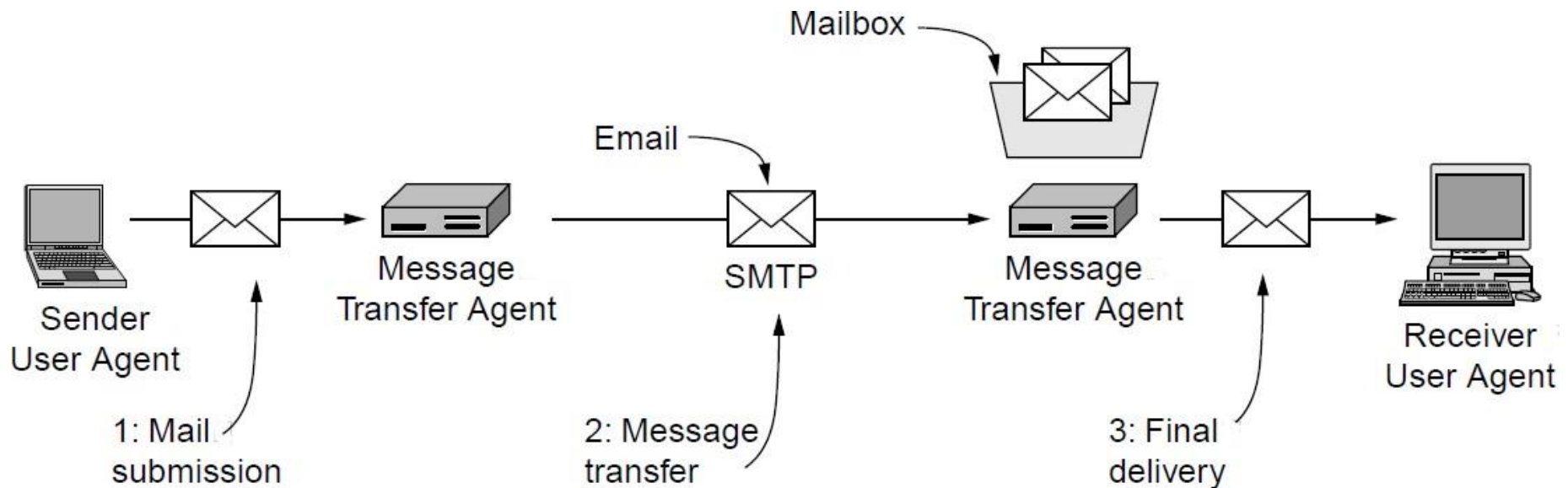
- Architecture and services »
- The user agent »
- Message formats »
- Message transfer »
- Final delivery »

Architecture and Services (2)



Architecture and Services (1)

The key components and steps (numbered) to send email



Architecture of the email system

1. User Agent (UA)

- A User Agent (UA) is an email program this allows users to **read, reply, search, filter, and delete** emails.
- UA Can include Gmail, Outlook, Thunderbird, Apple Mail.
- Can have graphical interfaces (modern systems) or text-based interfaces (older systems like Elm, mh, Pine).
- Can perform **automatic tasks** like spam filtering or vacation replies.
- Runs on the user's computer and operates only when the user is active.

What users see - interface elements of a typical user agent

Message folders

Mail Folders
All items
Inbox
Networks
Travel
Junk Mail

Message summary

From		Subject	Received
trudy	✉	Not all Trudys are nasty	Today
Andy	📎	Material on RFID privacy	Today
djw	!	Have you seen this?	Mar 4
Amy N. Wong		Request for information	Mar 3
guido		Re: Paper acceptance	Mar 3
lazowska		More on that	Mar 2
lazowska	📎	New report out	Mar 2
...		...	...

Search 🔍

Mailbox search

A. Student	Graduate studies?	Mar 1
Dear Professor, I recently completed my undergraduate studies with distinction at an excellent university. I will be visiting your ...		

Message

2. Message Transfer Agent (MTA)

- A **background process** that continuously runs on mail servers.
- Uses **SMTP (Simple Mail Transfer Protocol)** for transferring mail between servers.
- Ensures **automatic, reliable message delivery** and returns **delivery status or error reports**.
- Supports advanced features such as:
 - **Mailing lists**
 - **CC/BCC (carbon and blind copies)**
 - **Priority and encrypted mail**
 - **Alternate recipients and delegated access**

Message Formats (1)

Header fields related to message transport; headers are readable ASCII text

Header	Meaning
To:	Email address(es) of primary recipient(s)
Cc:	Email address(es) of secondary recipient(s)
Bcc:	Email address(es) for blind carbon copies
From:	Person or people who created the message
Sender:	Email address of the actual sender
Received:	Line added by each transfer agent along the route
Return-Path:	Can be used to identify a path back to the sender

Message Transfer (1)

Messages are transferred with SMTP (Simple Mail Transfer Protocol)

- Readable text commands
- Submission from user agent to MTA on port 587
- One MTA to the next MTA on port 25
- Other protocols for final delivery (IMAP, POP3)

SMTP

- **SMTP** is the **standard protocol** for **sending email** across the Internet.
- It uses **TCP port 25** to establish a reliable connection between the **sending** and **receiving** mail servers.
- The **receiving system** runs an SMTP server that:
 - Accepts incoming email messages.
 - Performs basic security checks.
 - Returns an **error report** if delivery fails.
- SMTP operates over **TCP**, ensuring reliable data delivery (no checksums needed).
- It is a simple ASCII-based protocol, making it: Easy to develop, test, and debug.

MIME (Multipurpose Internet Mail Extensions)

- **Before MIME:**

Early email (RFC 822) supported only **plain ASCII text**, mainly in English.

- **Problem:**

ASCII-only emails can be handle:

Accented characters (e.g., French, German)

Non-Latin scripts (e.g., Hebrew, Chinese)

Non-text content (e.g., images, audio, documents).

- **Solution:**

The **MIME standard** (RFCs 2045–2049) extended email to support **multimedia and international content** without changing the basic email protocols.

Message Formats (4)

Common MIME content types and subtypes

Type	Example subtypes	Description
text	plain, html, xml, css	Text in various formats
image	gif, jpeg, tiff	Pictures
audio	basic, mpeg, mp4	Sounds
video	mpeg, mp4, quicktime	Movies
model	vrml	3D model
application	octet-stream, pdf, javascript, zip	Data produced by applications
message	http, rfc822	Encapsulated message
multipart	mixed, alternative, parallel, digest	Combination of multiple types

Message Formats (3)

MIME header fields used to describe what content is in the body of the message

Header	Meaning
MIME-Version:	Identifies the MIME version
Content-Description:	Human-readable string telling what is in the message
Content-Id:	Unique identifier
Content-Transfer-Encoding:	How the body is wrapped for transmission
Content-Type:	Type and format of the content

3. Mailboxes and Final Delivery

- **Mailboxes** store incoming messages on mail servers.
- **User Agents** access and manage these mailboxes (view, delete, organize messages).
- The process of retrieving mail from the server is **called final delivery**.
- One mailbox can be accessed from multiple devices or user agents.

POP3, IMAP

POP3 (Post Office Protocol v3)

- POP3 **downloads** messages from the server to your device and usually **removes them from the server**.
- Changes made on one device (like deleting or reading emails) won't show up on others.

Ideal for: Single-device use or when you want emails stored **locally** (e.g., offline access).

- **IMAP (Internet Message Access Protocol)**
- Yes — IMAP keeps emails **on the server** and **synchronizes instantly** across all your devices.

Example: When you read or delete an email on your phone, the change appears immediately on your laptop too.

- People who use **multiple devices** (smartphone, tablet, PC).

The World Wide Web

- The Web, as the World Wide Web is popularly known, is an architectural framework for accessing linked content spread out over millions of machines all over the Internet

The World Wide Web

- The **World Wide Web (WWW)** is a system of interlinked hypertext documents accessed via the Internet. Users view and interact with web pages using a **web browser**.

Main Components

- The architecture of the WWW involves **three main components**:

a) Web Client (Browser)

Example: Chrome, Firefox, Edge.

Function: Sends requests to servers and displays the received web pages.

Uses **HTTP/HTTPS** to communicate.

b) Web Server

Example: Apache, Nginx, IIS.

Function: Stores, processes, and delivers web pages to clients.

Responds to HTTP requests from browsers.

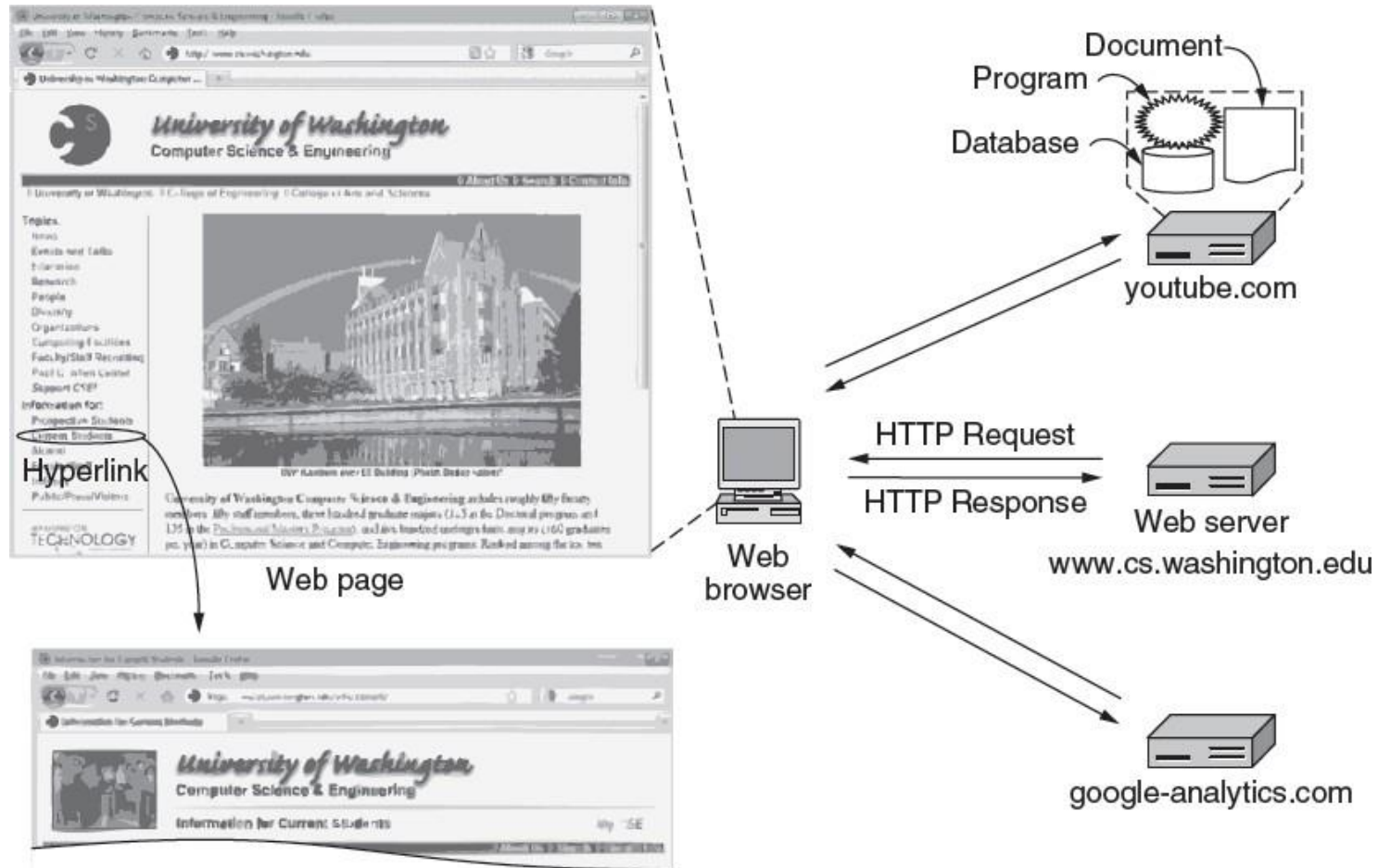
c) Web Resources

The actual content: HTML files, images, videos, CSS, JavaScript, etc.

Each resource has a **URL (Uniform Resource Locator)**.

Architectural Overview (1)

HTTP transfers pages from servers to browsers



Architectural Overview (3)

Steps a client (browser) takes to follow a hyperlink:

- Determine the protocol (HTTP)
- Ask DNS for the IP address of server
- Make a TCP connection to server
- Send request for the page; server sends it back
- Fetch other URLs as needed to display the page
- Close idle TCP connections

Steps a server takes to serve pages:

- Accept a TCP connection from client
- Get page request and map it to a resource (e.g., file name)
- Get the resource (e.g., file from disk)
- Send contents of the resource to the client.
- Release idle TCP connections

Architectural Overview (2)

Pages are named with URLs (Uniform Resource Locators)

- Example: <http://www.phdcomics.com/comics.php>


Protocol Server Page on server

Our
focus →

Name	Used for	Example
http	Hypertext (HTML)	http://www.ee.uwa.edu/~rob/
https	Hypertext with security	https://www.bank.com/accounts/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
mailto	Sending email	mailto:JohnUser@acm.org
rtsp	Streaming media	rtsp://youtube.com/montypython.mpg
sip	Multimedia calls	sip:eve@adversary.com
about	Browser information	about:plugins

Common URL protocols

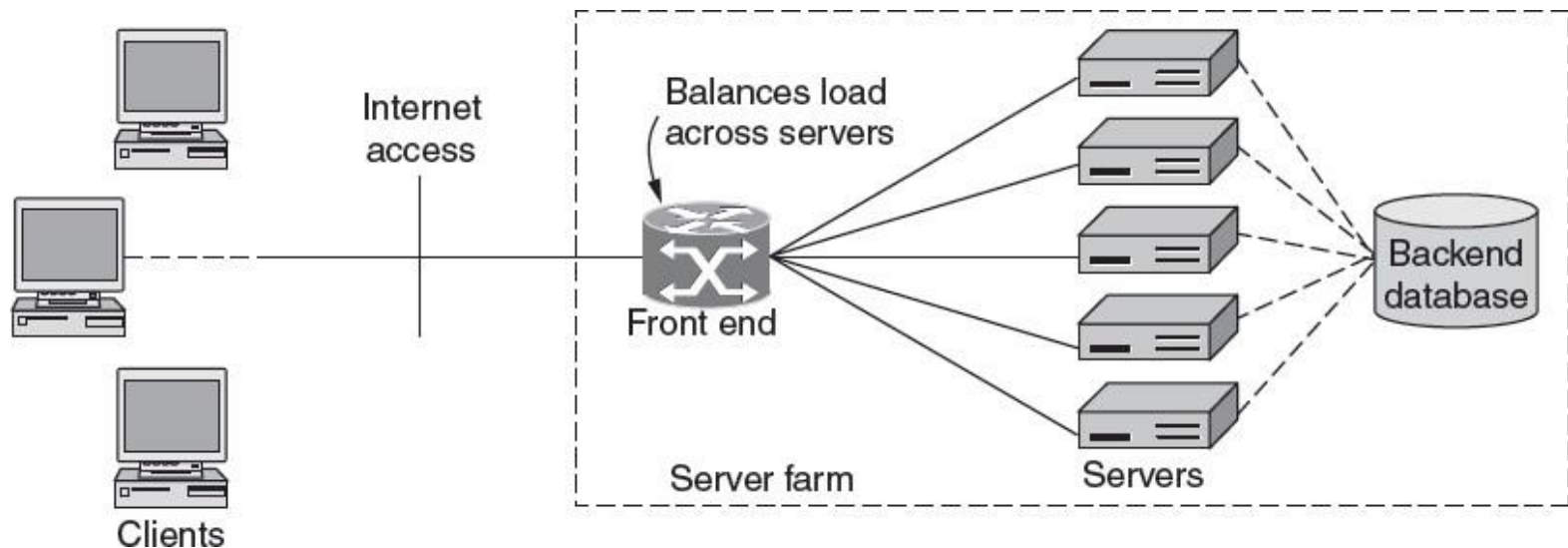
Content Delivery

Delivery of content, especially Web and video, to users is a major component of Internet traffic

- Server farms and Web proxies »
- Content delivery networks »
- Peer-to-peer networks »

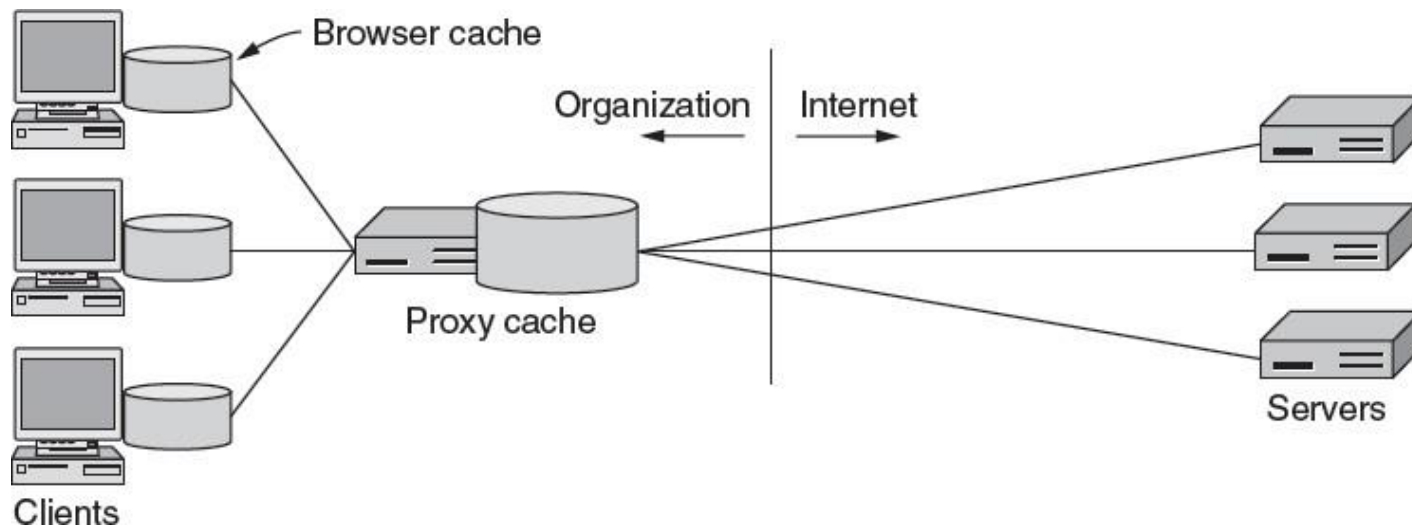
Server Farms and Web Proxies (1)

- A **server farm** is a group of servers located in one data center that store and process web applications and data.
- Front-end load-balances requests over servers
- Servers access the same backend database



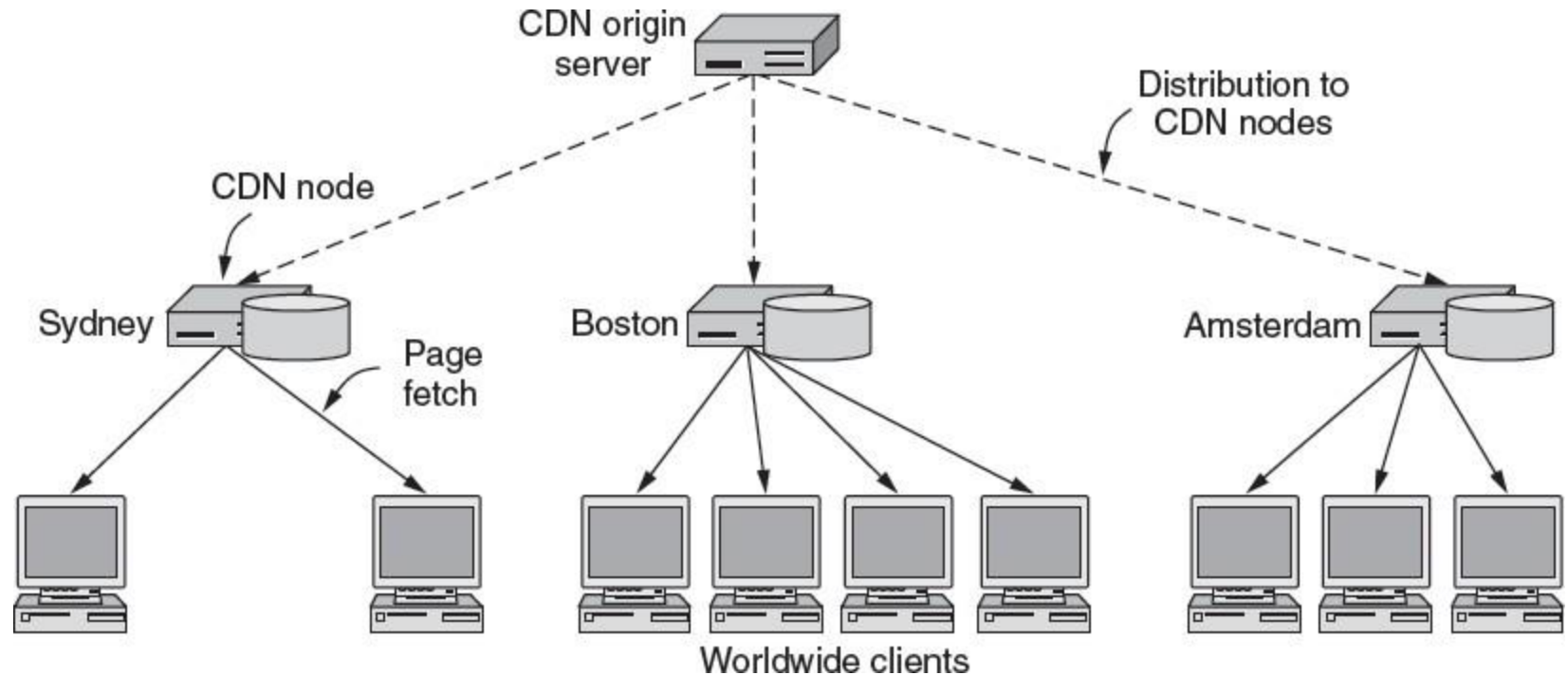
Web Proxies

- A web proxy is an intermediate server that caches and forwards requests between a client and a web server.
- When multiple users request the same content, the proxy can deliver it directly from its cache instead of fetching it again from the main server.



CDNs - Content Delivery Networks (1)

- A Content Delivery Network (CDN) is a distributed network of servers located in different geographical regions.
- CDNs deliver web and video content from the nearest server to the user to reduce delay and improve performance.



Content Delivery Networks (2)

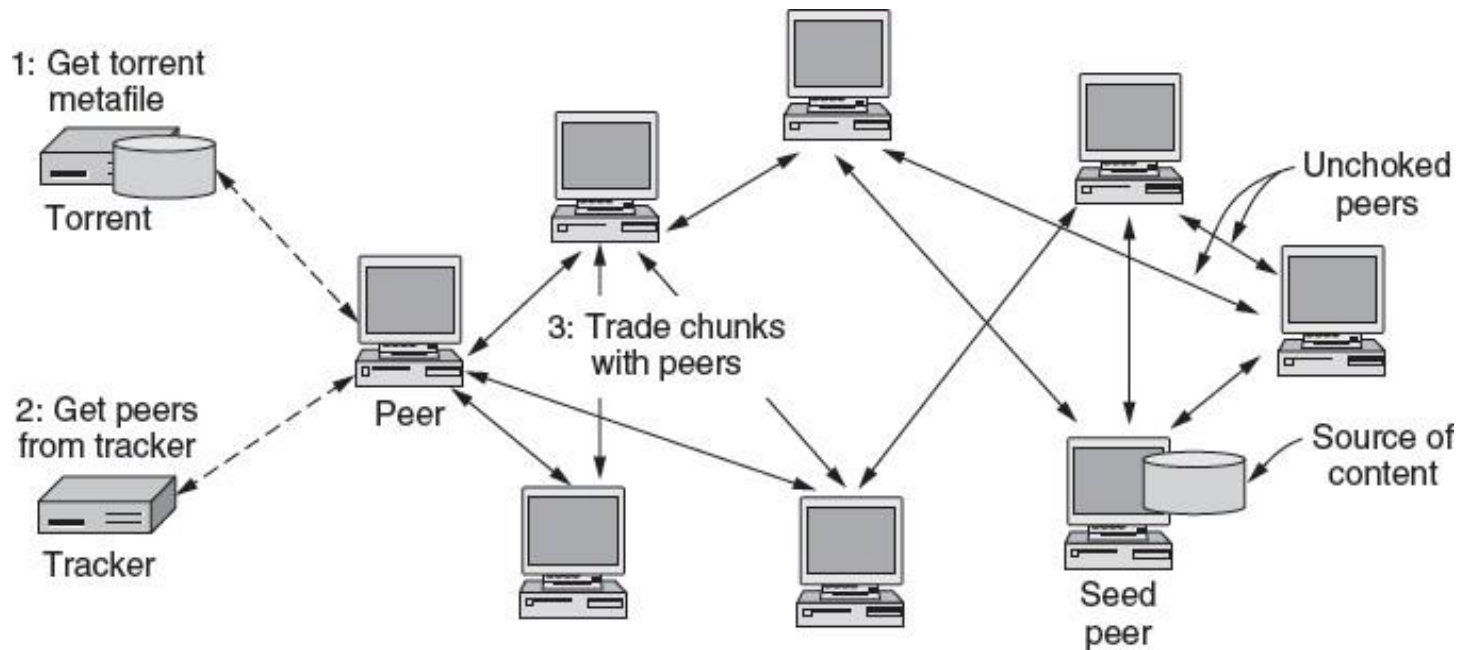
Directing clients to nearby CDN nodes with DNS:

- Client query returns local CDN node as response
- Local CDN node caches content for nearby clients and reduces load on the origin server

Peer-to-Peer Networks (1)

- A **Peer-to-Peer (P2P) Network** is a type of computer network where **each computer (peer)** acts as **both a client and a server**.

This means that every peer can **share its resources** (like files, storage, or bandwidth) **directly with other peers** — without needing a **central server**.



How It Works

- Each computer (peer) connects to other peers in the network.
- When a peer needs a file or data, it searches for other peers that have it.
- The file is **divided into small parts**, and each part is downloaded from multiple peers simultaneously.
- While downloading, the peer also uploads parts of the file to others — making sharing efficient and faster.

Applications

- Used to share large files directly between users.
- P2P connections allow players to **communicate directly** for faster gameplay.
- Some decentralized streaming platforms use P2P to distribute video efficiently.
- Cryptocurrencies like **Bitcoin** use P2P networks to share blockchain data among nodes.

Peer-to-Peer Networks (3)

DHTs (Distributed Hash Tables) because the basic functionality of an index is to map a key to a value. This is like a hash table, and the solutions are distributed versions, of course.

DHTs do their work by imposing a regular structure on the communication between the nodes, as we will see. This behavior is quite different than that of traditional P2P networks that use whatever connections peers happen to make. For this reason, DHTs are called structured P2P networks. Traditional P2P protocols build unstructured P2P network.

Peer-to-Peer Networks (3)

